

| Question  | Answer                                                                                                                                                                                         | Marks     |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 12(a)     | emission of particles/radiation by <u>unstable nucleus</u>                                                                                                                                     | <b>B1</b> |
|           | spontaneous emission                                                                                                                                                                           | <b>B1</b> |
| 12(b)(i)  | P – the curve that starts with a high number<br>D – the curve with the peak<br>S – the curve that increases from zero throughout<br><br><i>(one correct 1 mark, all three correct 2 marks)</i> | <b>B2</b> |
| 12(b)(ii) | $\lambda t_{1/2} = 0.693$<br><br>$\lambda = 0.693 / (60.0 \times 60)$                                                                                                                          | <b>C1</b> |
|           | $= 1.93 \times 10^{-4} \text{ s}^{-1}$                                                                                                                                                         | <b>A1</b> |
| 12(c)     | half-life of F is much shorter than half-life of E                                                                                                                                             | <b>B1</b> |
|           | <u>nuclei</u> of F decay (almost) as soon as they are produced                                                                                                                                 | <b>B1</b> |